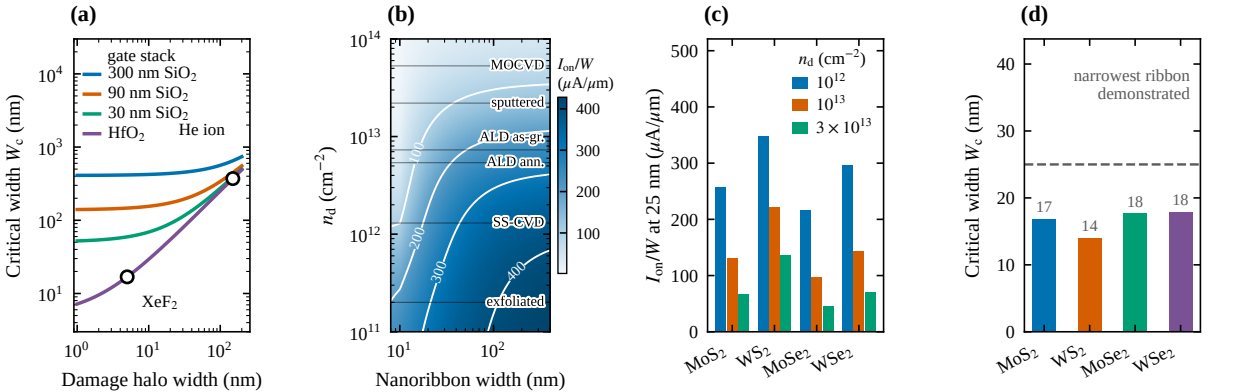

(a)
gate stack
300 nm SiO2
90 nm SiO2
30 nm SiO2
HfO2
He ion
XeF2
Critical width Wc (nm)
Damage halo width (nm)
(b)
MOCVD
sputtered
ALD as-gr.
ALD ann.
SS-CVD
exfoliated
100
200
300
400
Ion/W (μA/μm)
nd (cm−2)
Nanoribbon width (nm)
(c)
nd (cm−2)
10^12
10^13
3 × 10^13
Ion/W at 25 nm (μA/μm)
MoS2
WS2
MoSe2
WSe2
(d)
narrowest ribbon demonstrated
Critical width Wc (nm)
17
14
18
18
MoS2
WS2
MoSe2
WSe2